USER MANUAL

BARCC

Brain Atlas Regional Cell Counter

Version 8.03.000 - Generated June 05, 2026

BARCC is a specialized GUI application for analyzing immunofluorescence images. It enables researchers to overlay brain atlas sections, define regions of interest, perform automated cell counting, and export quantitative results.

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1. Introduction

BARCC (Brain Atlas Regional Cell Counter) is a specialized desktop application designed for researchers working with immunofluorescence (IF) microscopy images. It combines powerful image analysis tools with brain atlas registration to enable accurate, reproducible regional cell counting.

Purpose

The software allows users to overlay standardized atlas sections onto experimental TIFF images, define regions of interest through painting or atlas-based selection, apply sophisticated cell detection algorithms, and export quantitative results to Excel for further analysis.

Key Capabilities

- Import and display high-resolution TIFF images and multi-page PDF atlas files
- Interactive atlas alignment (translation, rotation, scaling)
- Freehand painting tools to define custom regions of interest
- Advanced, configurable cell detection with multiple thresholding methods
- Manual addition and removal of cells for quality control
- Automated cell counting within defined regions with Excel export
- Saving of annotated images and analysis sessions

NOTE

BARCC is particularly valuable for neuroscience studies involving c-Fos, NeuN, or other cell-type specific markers where regional quantification relative to brain anatomy is required.

What's New in Version 8.03.000

BARCC 8.03.000 delivers a major new workflow for atlas-based region editing via the Atlas Manager ribbon, plus global quick adjustments, improved mode visibility, and numerous robustness fixes for mixed image+atlas workflows.

- Atlas menu reorganization: "Import Atlas" (and global Crop/Move/Rotate/Scale plus per-region tools) now lives under a dedicated Atlas top-level menu (moved from File menu for better discoverability).
- New Atlas Manager ribbon (collapsible/expandable via header arrow; fully toggleable on/off from View > "Show Atlas Manager Ribbon" check item). The ribbon is the central hub for atlas region work:
- - Header always shows the currently selected region name/ID (or "No region selected").
- - Global tools now implemented as checkboxes (Crop, Move) so the active mode is immediately visible. Checking enables the corresponding click-drag behavior on the canvas (whole-atlas operations); unchecking returns to normal selection/naming.
- - "Move Selected Region" checkbox: when a region is selected (orange tint), click+drag inside it to translate *only* that zone's pixels in the mask. The underlying atlas artwork, other regions, and background image remain fixed. This is ideal for fine local alignment corrections without disturbing global registration.
- - "Border drag resize enabled" checkbox: when checked (and region selected), clicking near the perimeter of the orange/yellow region illuminates a local red edge segment. Re-click the red line to reposition the editable window; drag the red to locally push/pull only that portion of the boundary (linear falloff weights preserve connectivity and overall region integrity). Live shape update of the tinted region during drag via efficient rasterization + partial refresh. Release commits the edit.
- - Persistent red edge highlight survives panning, zooming, page changes, and show_page() calls (unless explicitly toggled off by a short click on an already-illuminated edge).
- - Quick Global Adjust section (new): Rot +5° / -5°, Scale +5% / -5% buttons that apply to the entire current atlas page (base artwork + all zone masks). Plus "Dialogs..." shortcut to the full global Rotate/Scale settings dialogs.
- - Selectable list of all labeled regions on the current atlas page. Clicking any entry selects it for editing (orange tint, ribbon header update, ready for quick adjust, edge drag, or move-selected). List auto-syncs after naming, transforms, page changes, etc.
- - Per-region quick adjust (unchanged from prior work but now clearly labeled "Selected Region Quick Adjust"): Rot +/-5°, Scale +/-5%, and "Dialogs..." for the currently selected region only (centroid-preserving transforms via the existing _apply_transform_to_region machinery).
- Automatic mutual exclusion between global modes and edge features: enabling the border-drag checkbox deselects global Move (and Crop); enabling global Move or Crop automatically unchecks the border drag. Region "Move Selected" also cooperates with edit mode bindings. This prevents confusing overlapping behaviors and makes the active toolset obvious from the checkbox states.
- Major improvements to per-region atlas editing: individual regions on an imported atlas can now be

selected (canvas click on named yellow area, list, or Atlas menu Select Region tool), then independently rotated, scaled, translated (via the move-selected drag), or locally deformed via the red-edge drag. The rest of the atlas and background stay in place. Transforms update both the visual tint and the underlying label mask used for counting.

- Robustness fixes across the board:

- - Atlas crop now correctly converts canvas rect to model (native) coordinates using _canvas_to_atlas, clamps to image bounds, rebases img_x/img_y so the cropped content stays visually in place, prunes orphaned zone names, and fully clears stale selection/edge state.

- - Load-order fixes: loading a TIFF/image *after* an atlas no longer leaves stale selected_zone_id or selected_edge_full_contour that would hijack global Move drags (via the priority checks in drag_start/drag_move) or cause is_near tests to behave unexpectedly. All image-load paths (import_tiff and file-browser _load_tiff_file) now perform complete selection/edge/region-mode clears in addition to the zone/mask wipes.

- - Edge grab hit-testing made forgiving: clicks that land on boundary pixels (zid==0 at exact integer sample) but are near the current selected region's border (or an already-illuminated red) are treated as edge-grab intent instead of falling through to a name prompt.

- - Global Move (edit_mode) and per-region features now correctly delegate in the drag handlers so that "Move Selected Region" or edge grab work even if the global Move binding is active (priority checks on mousedown; motion delegation).

- - Many additional state hygiene improvements (clears on page change, deselect, import atlas, crop end, etc.) so that ribbon list, orange tint, red edge, and move/edge flags stay consistent.

- Ribbon checkboxes (global Crop/Move + per-region Move Selected + Border drag) plus the selected region header and list now give immediate, at-a-glance feedback about the current editing context and which tools are armed.

- All new per-region and global quick adjust operations participate in undo (save_state) and correctly update the ribbon list/header after transforms.

These changes make fine-grained, region-by-region correction of atlas registration practical while preserving the ability to do global alignment. The visual distinction between "I'm moving the whole atlas" vs. "I'm only moving/deforming this one region" is now explicit via checkboxes and selection highlighting.

What's New in Version 8.02.002

BARCC 8.02.002 is a patch release that fixes a long-standing source of confusion between the counts the user sees visually on the annotated image/mask and the numbers reported in the spreadsheet.

- Fixed mismatch between per-zone cell counts shown on the mask/image (yellow zone labels like "RegionName\n(NN)" when "Show Zone Labels & Counts" is enabled, plus visible red cell markers in the annotated image) and the numbers in the exported spreadsheet (Cell Counts sheet) or CSV fallback.
- Root cause: The cell-to-zone assignment inside `count_cells_in_zones` (used for the DataFrame that feeds both the spreadsheet and `last_df`) and the on-screen label positioning in `show_page` unconditionally applied the atlas overlay offset (`-img_x` / `+ display_img_x` etc.).
- Painted regions (and the main experimental TIFF background) live in a 0,0-aligned coordinate system matching the cell detection mask. Atlas/PDF zones may come from a differently sized/positioned overlay layer. Any non-zero `img_x`/`img_y` (from panning the atlas, zooming, dragging, alignment, etc.) would shift the lookup, causing cells to be assigned to the wrong zone (or no zone) for counting purposes.
- Visual markers were drawn at the correct positions (using raw centroids), so users would see e.g. 30 red dots inside a painted region but the spreadsheet and labels would report a different number.
- Implemented a robust size-based heuristic:
- - Compare zone mask size to the main `background_image` / cell mask size.
- - If they match (within tolerance): zones are paint-on-background or standalone TIFF -> use direct cell coordinates (no offset) for accurate assignment and place labels over the background layer at canvas (0,0).
- - If sizes differ (atlas overlay): fall back to the previous `img_x`/`img_y` offset logic to align with the placed atlas layer.
- The same logic was applied to zone label text placement so the yellow (count) overlays appear over the correct regions even when an atlas overlay has a non-zero offset.
- After re-running Count Cells, the per-zone totals in the spreadsheet now exactly match the number of cells the user sees marked inside each region on the image and in the on-screen labels.
- No change to pure atlas workflows, detection logic, manual edits, or cases where no overlay offset is present. Version in exported settings JSON is "8.02.002".

This resolves the final major "numbers don't match what I see" issue for users relying on painted custom regions for quantitative analysis.

What's New in Version 8.02.001

BARCC 8.02.001 is a targeted patch release that resolves the last reported edge cases in the Paint tool for custom regions and improves dialog usability:

- First-paint reliability: Drawing a region, naming it immediately via right-click, and clicking Count Cells now succeeds on the *very first attempt* after loading any image (no more "No Regions Defined" for the initial named zone, with only later zones appearing in the spreadsheet).
- Fixed root cause in the 'img' bake path: The first `stop_paint()` (auto-called by Count Cells) + its `show_page()` would activate `atlas_filetype='img'` (via `save_paint`). `load_page_image()` then unconditionally reset `zone_names[page]`, `mask_images[page]`, and `zone_counters[page]` the first time `page_images` was populated for non-PDF content. This destroyed zones just registered by `name_painted_region` -> `_convert_named_paints_to_zones`. A guard now restricts the destructive per-page init to `atlas_filetype == 'pdf'` only; 'img' (baked paint) and 'png' (loaded paint) cases preserve existing zone/mask state.
- Additional conversion hardening (building on 8.02.000 durability work): broadened stroke collection in `_convert_named_paints_to_zones` (always tries `paint_group_data` model points + canvas + last-resort any paint items for still-named groups), conditional `dtag` only after successful retirement in `name_painted_region`, pre-clear re-try collection inside `stop_paint`, and an ultimate force-add of lingering data groups before the error guard in `count_cells`.
- Brightness slider dialog X button fix: The titlebar X (and Alt+F4) on the Brightness Settings window (containing the live brightness scale/slider) now properly closes the dialog. Previously wired to a no-op `disable_event`. Applied the same `window.protocol("WM_DELETE_WINDOW", window.destroy)` pattern (already used by Mask Settings) to Brush Settings, Scale Settings, and Rotate Settings for consistency. Transparent window registration and <Destroy> cleanup continue to work. Progress dialogs remain hardened (defensive no-op flag) so early close cannot crash counting/detection.
- All prior v8.02.000 guarantees (immediate naming registers zones, Count auto-stops paint, durable model coordinates, `binary_fill_holes` + neighborhood lookup for accurate interiors, retirement to prevent dups, full wipe on every new image load, auto Save Paint Layer into the left File Browser directory, etc.) now apply reliably even to the first painted+named region on a fresh load.

These changes complete the Paint tool reliability story. The workflow "draw, right-click name immediately after the stroke, click Count Cells (without pressing Stop Paint)" is now fully production-ready on the first try.

What's New in Version 8.02.000

BARCC 8.02.000 delivers major stability, accuracy, and reliability improvements focused on the Paint tool workflow for creating custom analysis regions:

- Painted zones that are named immediately after drawing (via right-click) now correctly and reliably register for cell counting. Clicking Count Cells automatically stops paint mode and converts all strokes - both user-named regions and auto-defaulted "Painted Region N" entries.
- Complete prevention of cross-image contamination: every new TIFF loaded (via File Browser or Import) performs a full wipe of all mask_images, zone_names, zone_counters, named_paint_groups, and durable paint geometry data.
- Durable storage of paint stroke geometry using stable model/image coordinates (not view-dependent canvas coords). Hand-drawn regions now survive zooming, panning, and internal canvas refreshes.
- Accurate interior filling of painted structures using scipy.ndimage.binary_fill_holes after boundary stroking. Cells that are visibly inside your drawn outlines are now counted correctly instead of only thin boundary pixels.
- Neighborhood search (5x5) around each detected cell centroid when looking up zone membership. This provides tolerance for drawing precision and any remaining micro-imperfections in region fills.
- Critical stability fix: Closing the "Counting Cells" or "Detecting Cells" progress dialog with the X button (or Alt+F4) before the operation finishes no longer crashes the application. All progress UI methods are now fully defensive no-op after early dismissal.
- Elimination of duplicate zone entries (e.g., 6 zones appearing when only 3 were drawn) through proper retirement of processed paint groups after conversion.
- Overall hardening of the entire paint -> named group -> zone mask -> counting pipeline.
- Paint menu reorganization: "Save Paint" moved from File menu to Paint menu and renamed "Save Paint Layer". New "Load Paint" command added.
- Save Paint Layer now auto-saves directly into the folder currently open in the left File Browser (with automatic unique naming to avoid overwrites), matching the auto-export style of Count Cells. No save dialog appears when a working directory is selected.
- Load Paint and Import Paint now default their file dialogs to the current directory shown in the left File Browser for faster workflow.

These changes make the Paint tool (for custom regions outside of atlas PDFs) fully trustworthy and production-ready for quantitative cell counting work.

What's New in Version 8.01.000

BARCC 8.01 introduces major improvements to the cell detection system and parameter tuning workflow:

- New default detection engine based on Laplacian-of-Gaussian (blob_log) blob detection. This method is significantly more robust on typical immunofluorescence images than the previous watershed approach.
- Full set of Blob Detection parameters now exposed in Mask Settings with live preview.
- New "Smart Suggest (Offline)" button - a completely local analysis tool that examines your image and current detections and recommends better parameter values. No data is ever transmitted.
- Ability to instantly switch between the new Blob method and the legacy Watershed method.
- Autotune buttons now intelligently adapt their adjustments depending on the active detection method.
- When using Add Cell or Remove Cell, the Brush Settings dialog now opens automatically for immediate size control.

These changes dramatically improve the experience of tuning detection parameters for difficult or variable images.

2. Installation & Requirements

System Requirements

- Windows 10 or later (primary supported platform)
- Python 3.8 or higher
- At least 8 GB RAM recommended for large images

Image Format and Compatibility Requirements

BARCC is designed primarily for quantitative analysis of immunofluorescence images. The following specifications describe the supported and recommended image characteristics:

Supported File Formats

BARCC supports the following image file formats with varying levels of compatibility:

- TIFF (.tif, .tiff) - Fully supported and strongly recommended. Supports both single-page and multi-page (stacked) TIFF files. Handles 8-bit, 16-bit, and 32-bit floating point data. This is the only format recommended for serious quantitative work.
- PNG - Limited support. May be imported in certain workflows but is not recommended for primary analysis due to potential compression artifacts.
- JPEG / JPG - Not supported. Lossy compression makes these formats unsuitable for cell detection and quantitative analysis.

Bit Depth and Data Types

- 8-bit (uint8) - Supported
- 16-bit (uint16) - Fully supported and recommended for most immunofluorescence work
- 32-bit floating point - Supported (both 0-1 normalized and arbitrary ranges)
- Multi-channel images (RGB, RGBA, etc.) - Accepted but typically converted or reduced internally. Single-channel grayscale images are strongly preferred for best results.

Image Dimensions and Size

BARCC has no hard upper limits on image dimensions, but practical performance considerations apply:

- Recommended maximum: approximately 8,000-10,000 pixels on the longest side for comfortable interactive use.
- Very large images (e.g., > 12,000-15,000 pixels per side) may cause high memory usage, slow performance during detection, or instability during manual editing.

- File sizes above ~500 MB can lead to noticeable slowdowns in the Mask Settings dialog and when generating visualizations.

Resolution and Physical Scaling

BARCC is strictly pixel-based and does not read or use image resolution metadata (DPI or µm/pixel). All detection parameters (cell size, sigma values, peak intensity, etc.) are expressed in pixels. Users must convert biological size requirements into pixel values based on their specific imaging resolution.

Recommended Image Characteristics

For optimal performance with the current detection system (especially the Blob Detection method):

- File format: Uncompressed or lossless TIFF
- Bit depth: 16-bit or 32-bit floating point
- Background: Relatively dark and uniform
- Cell appearance: Locally brighter than background, roughly circular or elliptical
- Noise level: Moderate to low (use preprocessing denoising when necessary)
- Image size: 1,000-8,000 pixels on the longest side for best interactive experience

NOTE

While BARCC can technically load images outside these recommendations, results may be suboptimal. For best accuracy and performance, prepare images as single-channel TIFFs with good contrast between cells and background.

Installation Steps

1. Clone the repository:

```
git clone https://github.com/LaingLab/BARCC.git
cd BARCC
```

2. Install dependencies:

```
pip install -r requirements.txt
```

For full .xlsx export support (including the Detection Parameters metadata sheet) and the _masked.tif feature, also install the Excel engines:

```
pip install openpyxl xlsxwriter
```

For the best experience with automatic Excel exports (including a second sheet with all detection parameters), we also recommend installing the Excel engines:

```
pip install openpyxl xlsxwriter
```

If these packages are missing, BARCC will automatically fall back to saving results as a plain .csv file instead of .xlsx. These packages are listed as recommended (but not strictly required) in the project's requirements.txt.

3. Launch the application:

```
cd Application
python barcc.py
```

3. Getting Started

Upon launching BARCC you are presented with a large central canvas area and a menu bar across the top. The typical workflow follows these steps:

- Import a TIFF image (File > Import TIFF)
- Import an atlas section PDF (Atlas > Import Atlas)
- Align the atlas over your image using Move, Rotate, and Resize tools
- Define regions using either Paint tools or by clicking atlas regions
- Configure and preview cell detection parameters (Mask > Show Mask Settings)
- Run cell counting and review results
- Export data to Excel and save annotated images

4. User Interface Overview

The main interface consists of a left file browser pane and a large central image canvas. The left pane lets you select a folder and browse all TIFF images within it. Double-clicking any file loads it as the active image. A checkmark column shows which images have already been counted (based on the presence of matching .csv or .xlsx result files).

All other functionality is accessed through the top menu bar. The interface is designed to keep as much screen space as possible available for the image and mask visualization.

Main Canvas

The large central area displays your experimental image with the atlas overlay on top (when loaded). You can pan using the scrollbars or Alt+drag. Use the mouse wheel to zoom in and out. Zoom is centered on the mouse cursor and keeps painted regions, atlas overlays, and masks perfectly aligned with the background image at all zoom levels.

Menus

File, Edit, Atlas, Paint, Mask, and Cell menus provide access to all features. Many operations (especially in Mask Settings) open auxiliary dialogs for parameter adjustment.

5. File Menu

Import TIFF

Loads the primary experimental image. Supported formats include single and multi-page TIFFs. After import, the image is automatically scaled to fit the window while preserving aspect ratio.

Import Atlas

Opens a PDF file containing brain atlas plates. BARCC uses PyMuPDF to render individual pages at high quality. You can navigate between pages of multi-plate PDFs.

Note: Import Atlas (and related global/per-region atlas tools) now lives under the top-level Atlas menu rather than the File menu, for better grouping with Crop, Move, Rotate, Scale, and the new region selection/editing commands.

Split Tiff

Utility for breaking apart stacked (multi-page) TIFF files into individual images. Useful when your source data contains multiple sections in one file.

File Browser (Left Pane)

BARCC v8.01 introduced a dedicated file manager pane on the left side of the main window. This makes it much easier to work with folders containing many TIFF images.

Click the "Select Folder" button at the top of the left pane to choose a directory. BARCC will scan the folder and display all .tif and .tiff files in a list. Double-click any file in the list to load it as the active working image.

A second column in the list displays a checkmark ([x]) next to any image that has already been processed. This checkmark appears automatically if a matching .csv or .xlsx results file (generated by Count Cells) exists in the same folder. This is very useful for tracking which images in a large dataset have already been counted.

The Refresh button rescans the current folder. This is useful if you add or remove files while BARCC is running.

NOTE

> The File Browser works independently of the traditional "File > Import TIFF" menu. You can still use the menu for one-off files, but the left pane is much faster when working with a whole folder of images.

Save Flattened Image

Exports a composite image containing both the original TIFF and the currently aligned atlas as a single JPEG. This is useful for figure preparation and record keeping.

Save Paint

Saves freehand painted regions to a reusable file format for later sessions.

6. Working with Atlas Sections

Accurate alignment of the atlas to your experimental image is critical for meaningful regional analysis. BARCC provides several interactive tools for this purpose, now centered around the Atlas Manager ribbon for both global and per-region fine control.

The Atlas Manager Ribbon

The ribbon (View > "Show Atlas Manager Ribbon" to toggle visibility) is the primary interface for working with atlas regions. It is collapsible via the >/v arrow in the header.

- Header always displays the currently selected region (name and ID) or "No region selected".
- Global tools appear as checkboxes (Crop, Move) - their checked state shows at a glance whether whole-atlas click-drag modes are active.
- "Move Selected Region" checkbox: enables interior drag of the orange-tinted selected region to shift *only* its mask pixels (everything else stays fixed).
- "Border drag resize enabled" checkbox: arms edge-grab mode for the selected region.
- Global Quick Adjust: one-click Rot +/-5° and Scale +/-5% for the entire current atlas page (base + all masks), plus quick access to the full Rotate/Scale dialogs.
- Selectable list of every labeled region on the current page. Click any entry to select it for editing (orange highlight appears, ribbon header updates).
- Selected Region Quick Adjust: the same +/- rotate and scale buttons, but applied only to the currently selected region (centroid-preserving).
- All operations are undoable and immediately reflected in the ribbon list and on-screen tints.

NOTE

The ribbon automatically stays in sync after naming, transforms, page changes, crops, etc. Checkboxes enforce mutual exclusion (e.g., turning on border drag automatically unchecks global Move and vice versa) so you always know which tool family is armed.

Selecting Regions for Editing

Click a yellow (named) or orange (selected) area directly on the atlas canvas, or use the list in the ribbon, or Atlas menu > "Select Region" (temporarily rebinds clicks to pick a zone). The selected region receives an orange tint overlay and becomes the target for quick adjust, edge drag, and move-selected.

Canvas clicks on already-named regions now intelligently autoselect them into the ribbon (instead of re-prompting for a name). Clicks near the perimeter (even on boundary pixels) are treated as edge-grab intent when the border checkbox is on.

Global Quick Adjust (new in 8.03)

The ribbon now offers the same style of quick +/- buttons for the *entire* atlas page that were previously available only for individual selected regions:

- Rot +5° / Rot -5° - rotates the full rendered atlas artwork and all zone masks for the current page (expand=True semantics, size may grow).
- Scale +5% / Scale -5% - uniform scaling of the entire page content.
- "Dialogs..." shortcut to the full global Rotate and Scale settings dialogs (with numeric entry and axis-specific scaling).

These are the global equivalents of the per-region quick adjust. They affect every region on the page plus the underlying artwork; use them for coarse global corrections before fine per-region work.

Per-Region Editing & Quick Adjust

Once a region is selected (orange), you can:

- Use the Selected Region Quick Adjust buttons (Rot +/-5°, Scale +/-5%) - these transform only the chosen zone's mask while keeping its centroid roughly in place. The rest of the atlas and background are untouched.
- Open full per-region Rotate/Scale dialogs via the "Dialogs..." button (or Atlas menu).
- Enable "Move Selected Region" and drag inside the orange area to translate only that zone (see below).
- Enable "Border drag resize" for local boundary editing (see below).

Move Selected Region (translate only one zone)

With a region selected and the "Move Selected Region" checkbox checked, click and drag inside the orange area. Only the pixels belonging to that zone in the label mask are shifted; the atlas artwork itself, other zones, and the background image remain exactly where they are. This is perfect for nudging a single misaligned anatomical region without disturbing your global registration.

The checkbox and the global Move checkbox are coordinated so you can easily switch between moving the whole atlas layer versus moving just the current region.

Edge Grab & Local Deformation (expand/shrink from the border)

With "Border drag resize enabled" checked and a region selected:

- Click near the perimeter of the orange/yellow region. A local segment of the boundary is highlighted in bright red.
- The red segment is a movable "handle" centered on your click point (using a sliding window along the contour with falloff weighting).

- Click again on an already-red edge to re-center the editable window at that exact location.
- Drag the red line outward or inward. Only the vertices inside the local window move, with linear falloff so the deformation blends smoothly into the rest of the boundary. The rest of the region (and the rest of the atlas) stays fixed.
- You get live visual feedback: the tinted region shape updates in real time as you drag.
- Release the mouse to commit the new shape to the mask (used for counting) and the display.
- A short click (no drag) on a red edge toggles the highlight off.

The red highlight is persistent across pans, zooms, and show_page refreshes until you explicitly dismiss it. This gives you precise, one-sided control over region boundaries without having to repaint or globally scale/rotate.

Global Crop & Move (checkbox style)

The Global Crop and Move tools are now checkboxes in the ribbon (and check items in the Atlas menu). When checked, they rebind canvas clicks for whole-atlas operations (crop rectangle or drag-to-move the entire overlay layer via img_x/img_y). Their checked state gives clear feedback that you are in a global mode rather than a per-region editing mode. Unchecking (or enabling an edge feature) restores normal click behavior (naming / region selection / edge grab).

NOTE

Best practice for atlas work: Use global Crop/Move/Quick Rotate/Scale first for rough alignment of the whole plate, then switch to per-region tools (list or canvas selection + quick adjust + edge drag + move-selected) for fine anatomical corrections. The checkboxes and orange selection tint make the current context obvious.

Move Atlas (legacy / still available)

The classic click-and-drag of the atlas overlay (when the global Move checkbox is checked) repositions the entire atlas layer relative to the background. This remains the primary tool for coarse global alignment.

Rotate / Scale (global)

Use either the ribbon Global Quick Adjust buttons for small increments or the full dialogs (Atlas menu or ribbon "Dialogs..." under Global Quick Adjust). These affect the rendered atlas artwork and all zone masks on the current page. After a global rotate that expands the canvas, the layer offset is automatically handled on the next redraw.

Brightness & Contrast Adjustments

The atlas rendering can be lightened or darkened independently of the experimental image to improve visibility of boundaries during alignment.

NOTE

Best practice: Identify 3-4 reliable anatomical landmarks (e.g., ventricles, major fiber tracts, cortical boundaries) and align to those rather than trying to match the entire section at once. After global alignment, use the new per-region tools in the ribbon to tweak individual structures without disturbing the rest of the plate.

7. Paint Tools for Regions of Interest

When the standard atlas regions do not match your experimental needs, the Paint tools allow completely custom region definition.

Start / Stop Paint

Activates freehand drawing mode. Draw directly on the image with the mouse.

Pen vs Eraser

Switch between adding area (Pen) and removing area (Eraser). Brush size is adjustable via the Brushsize dialog.

Load / Save Paint Layer

Painted regions can be saved and reloaded in future sessions, enabling consistent analysis across multiple images or experiments.

The Paint menu now contains dedicated **Load Paint** and **Save Paint Layer** commands. When a folder is open in the left File Browser, Save Paint Layer automatically saves the current paint layer directly into that folder as `{image}_paint.png` (or with numbered suffixes to avoid overwriting). The left file list is refreshed automatically after saving. Both Load Paint and Import Paint default their file selection dialogs to the current working directory shown in the left File Browser.

Labeling Painted Regions

While in Paint mode (before clicking Stop Paint), you can right-click on any painted stroke to name it. BARCC treats each continuous drawing action (mouse button down through release) as a single "structural boundary." All connected line segments belonging to that stroke are grouped together.

When you right-click a stroke:

- The entire connected group is highlighted in yellow for visual feedback.
- A dialog appears allowing you to enter or edit a name for the region.
- Named painted regions are converted into proper analysis zones when you click Stop Paint.
- These named regions participate in cell counting exactly like atlas regions and will appear in your results with the names you assigned.

NOTE

You can only label painted regions while the Paint mode is active (i.e., before you click "Stop Paint"). Once you stop painting, the strokes are committed to the persistent paint layer and the live canvas items used for naming are cleaned up.

8. Mask Settings & Cell Detection

This is the most powerful and configurable part of BARCC. The Mask Settings dialog provides fine-grained control over cell detection. BARCC v8.01+ supports two different detection strategies that you can switch between at any time:

- Blob Detection (Recommended): Uses modern Laplacian-of-Gaussian (blob_log) blob detection. Generally provides the best results on immunofluorescence images with variably bright cells.
- Watershed (Legacy): The original threshold + distance transform + watershed pipeline. Retained for compatibility with older workflows.

You can switch between these two methods at any time using the radio buttons at the bottom of the Mask Settings dialog. Most users should use the Blob method for new work.

NOTE

A powerful new feature in v8.01 is the "Smart Suggest (Offline)" button. This fully local tool analyzes your current image and detection results and suggests better parameter values. No data ever leaves your computer.

Detection Method

At the bottom of the Mask Settings dialog you will find a clear choice between two detection engines:

- Blob (new/recommended): Modern multi-scale blob detection using Laplacian of Gaussian. Much more robust for typical fluorescent cell images.
- Watershed (legacy): The original method based on global/local thresholding followed by watershed segmentation.

When Blob is selected, the lower part of the dialog shows the Blob Detection parameters. When Watershed is selected, the legacy Watershed parameters are shown instead.

Blob Detection Parameters (Recommended)

These parameters control the modern blob-based detector:

Blob Threshold

The most important sensitivity control. Lower values detect more (and dimmer) cells but increase false positives. Higher values are more conservative. Typical useful range: 0.05 - 0.20.

Blob Min / Max Sigma

Controls the range of cell sizes the detector will look for. Min Sigma sets the smallest detectable feature size;

Max Sigma sets the largest. For most immunofluorescence nuclei, Min Sigma around 1.5-3.0 and Max Sigma around 7-12 works well.

Blob Num Sigma

Number of scales tested between Min and Max Sigma. Higher values give finer granularity at the cost of speed. Default (12) is usually sufficient.

Blob Overlap

How much overlap is allowed between nearby blobs. Lower values reduce duplicate detections on clustered cells.

Blob Min / Max Area

Post-detection filters based on area (in pixels). Very effective at removing tiny noise blobs or huge clumps.

Blob Min Circularity

Requires detected blobs to be reasonably round. Raising this value helps reject irregular artifacts.

Threshold Methods

The Threshold Method controls how the image is converted to a binary (black and white) mask for cell detection. Four options are available:

- Otsu: Automatically determines the optimal global threshold using Otsu's method. Fast and effective on images with good contrast.
- Adaptive: Uses local areas to determine the threshold. Excellent for images with varying brightness across the field of view.
- Local: Similar to Adaptive but uses a different neighborhood computation. Useful when Adaptive produces too many or too few detections.
- Manual: Uses a fixed threshold value (0.0-1.0) that you specify. Provides maximum reproducibility across batches of images.

Cell Detection Parameters

These parameters control how individual cells are identified from the binary mask:

Manual Threshold

Range: 0.0 to 1.0. Only used when Threshold Method is "manual". Higher values make detection more selective (fewer cells). Lower values increase sensitivity.

Adaptive Block Size

Must be an odd integer (e.g. 51, 101, 151). Size of the local region used for adaptive thresholding. Larger values consider more surrounding area. Smaller values are more sensitive to local changes. Recommended range: 51-151 pixels.

Local Radius

Integer value (typically 5-30). Size of the neighborhood for the Local threshold method. Larger values smooth noise but may miss smaller cells.

Min Cell Size / Max Cell Size

Integer values in pixels. Define the acceptable size range for an object to be counted as a cell. Min Cell Size filters noise and small artifacts. Max Cell Size excludes clumps. Typical values depend on magnification (e.g. Min 20, Max 100-200).

Circularity Threshold

Range: 0.0 to 1.0. How circular a region must be to be accepted as a cell (1.0 = perfect circle). Higher values enforce rounder shapes. Typical value: 0.7.

Min Peak Distance

Integer value (pixels). Minimum distance required between detected cell centers. Helps prevent over-counting of touching cells. Typical values: 5-10 pixels.

Peak Min Intensity

Range: 0.0 to 1.0. Minimum brightness required for a local maximum to be considered a cell center. Higher values detect only brighter cells. Lower values detect dimmer cells. Typical starting value: 0.1.

Watershed Compactness

Range: 0.0 to 1.0. Controls how the watershed algorithm separates touching cells. Higher values favor more compact, rounder boundaries. Lower values follow intensity gradients more closely. Typical value: 0.0.

Base Multiplier & Sensitivity Range

Advanced sensitivity controls. Base Multiplier sets overall detection sensitivity (default ~1.1). Sensitivity Range controls how much the sensitivity slider can influence results (default 0.2).

Preprocessing Pipeline

A full preprocessing chain can be applied before cell detection. Each step can be enabled or disabled independently. The dialog only shows parameters relevant to the selected methods.

Background Method

Options: tophat or none. Controls removal of large-scale background variations. Tophat uses morphological operations and is generally recommended. When enabled, the Ball Radius (structural element size, default 15) controls how large-scale the background variations removed are.

Denoise Method

Options: gaussian, median, bilateral, or none. Reduces noise before detection.

- Gaussian: Applies a Gaussian blur (controlled by Gaussian Sigma, typically 0.1-5.0).
- Median: Better at preserving edges (controlled by Median Kernel size).
- Bilateral: Preserves edges while smoothing (controlled by Bilateral Sigma Color and Bilateral Sigma Space).

Contrast Method

Options: stretch, clahe, gamma, or none. Enhances contrast before detection.

- Stretch: Simple linear contrast stretching.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): Local contrast enhancement. Controlled by CLAHE Kernel (typically 8-16) and CLAHE Clip Limit (typically 1.0-4.0).
- Gamma: Gamma correction. Values < 1 brighten the image; values > 1 darken it.

Enhance Method

Currently supports Unsharp Mask sharpening. Controlled by Unsharp Radius (typically 0.1-5.0) and Unsharp Amount (strength of sharpening, typically 0.1-5.0). Useful for making cell boundaries crisper.

Both manual mask editing overlays (red) and automatic cell detection masks remain visible and correctly aligned when you zoom in or out using the mouse wheel.

NOTE

Start with the default settings and adjust one parameter at a time while using "Show Mask" to preview the binary detection result. This iterative approach is the fastest way to obtain high-quality cell counts.

Autotune Panel

The Mask Settings dialog includes a convenient Autotune panel with one-click adjustments. These buttons intelligently adapt their behavior depending on whether you are using the Blob or Watershed detection method.

The available Autotune buttons are:

- More cells - Increases overall sensitivity. With Blob mode this primarily lowers the Blob Threshold and Min Sigma.

- Less cells - Decreases sensitivity and raises size/shape requirements.
- Bigger cells / Smaller cells - Adjust size-related parameters (Min/Max Area or Min/Max Cell Size).
- Brighter cells / Dimmer cells - Primarily adjust intensity sensitivity (Blob Threshold or Peak Min Intensity).

Note: The Autotune buttons are intentionally conservative. For best results on difficult images, use the new "Smart Suggest (Offline)" button instead (see below).

Export / Import Settings

You can now export your current detection and preprocessing settings as a portable .json file. This is useful for backing up configurations, sharing them with colleagues, or moving them between computers.

In the Mask Settings dialog, use the buttons at the bottom:

- "Export Settings..." - Saves your current Blob (or Watershed) configuration and preprocessing settings to a .json file of your choice.
- "Import Settings..." - Loads a previously exported .json file and applies all parameters immediately.

This is separate from the internal Presets system (which stores quick named presets locally in ~/.barc/presets.json).

Smart Suggest (Offline) - New in v8.01

This is one of the most powerful new features in BARCC 8.01. Clicking "Smart Suggest (Offline)" runs a fully local analysis on your current image and detection results. It then proposes specific parameter improvements with clear explanations for each suggestion.

- Everything runs 100% on your computer - no images or data are sent anywhere.
- Each suggestion has a checkbox. You can selectively choose which changes to apply.
- Buttons at the bottom allow you to "Apply All", "Apply All That Are Checked", or simply "Close".
- It works with both Blob and Watershed modes and gives context-aware advice based on your actual data.

This tool is especially useful when the simple Autotune buttons are too aggressive or not aggressive enough. It is the recommended way to get good starting parameters for new or difficult images.

9. Manual Cell Editing

No automated detector is perfect. BARCC provides direct manual editing tools so you can correct errors before counting.

Add / Remove Cells

- Add Cell - Click or paint to mark additional cells that the detector missed.
- Remove Cell - Erase false positives (dust, autofluorescence, etc.).
- Brush Size - When you activate Add Cell or Remove Cell, the Brush Settings dialog now opens automatically so you can immediately adjust dot size.

Edits are applied to a separate overlay mask and can be toggled on and off for comparison. All manual edits are included in the final cell count.

10. Counting Cells & Exporting Results

Once regions are defined and detection parameters are tuned, click "Count Cells" under the Cell menu.

BARCC will compute the number of detected cells within each named region. Results are now saved **automatically** (no file dialog) with the following files created in the same folder as your source TIFF:

- `YourImage.xlsx` - Excel workbook with two sheets:
- - Cell Counts - Region name, cell count, area, density, etc.
- - Detection Parameters - Complete record of every setting used (both cell detection and preprocessing). This is extremely useful for reproducibility and methods sections.
- `YourImage_masked.tif` - The original image with the final cell mask (including any manual Add/Remove edits) drawn as a semi-transparent red overlay. Ready for figures or further analysis.

BARCC automatically saves two files when you click Count Cells (no manual Save dialog):

- `YourImage.xlsx` - Contains two sheets: "Cell Counts" (the actual results) and "Detection Parameters" (a complete record of every setting used for reproducibility).
- `YourImage_masked.tif` - The original image with the final cell mask (after all manual Add/Remove edits) drawn as a semi-transparent red overlay. This is very useful for figure preparation.

To generate the .xlsx file (instead of falling back to .csv), the following packages are required:

```
pip install openpyxl xlsxwriter
```

These are listed as recommended (but not strictly required) in the project's requirements.txt. Without them, results are saved as a plain CSV file.

11. Saving & Export Options

- Flattened Image (JPEG) - Final figure-ready composite of image + atlas + annotations
- Paint files - Reusable region definitions
- Automatic Excel export on Count Cells (with full Detection Parameters metadata sheet)
- Automatic _masked.tif export (original image + final cell mask overlay)

12. Keyboard Shortcuts

Shortcut	Action
Ctrl + Z	Undo last action (move, rotate, paint, highlight, etc.)
Ctrl + S	Save flattened image

13. Troubleshooting

Common Issues

- Cells not detected: Try lowering Peak Min Intensity or switching to Adaptive threshold.

- Too many false positives: Increase Min Cell Size and Circularity Threshold.

- Atlas looks blurry: Re-render at higher zoom or adjust brightness settings.

- Performance is slow: Close other applications; very large images benefit from 16 GB+ RAM.

Getting Help

For bugs or feature requests, please open an issue on the GitHub repository:
https://github.com/LaingLab/BARCC

Thank you for using BARCC

We hope this software accelerates your research. Feedback and contributions are always welcome.